

Project Initialization and Planning Phase

Date	15 March 2026
Team ID	SWTID-2026-4683
Project Title	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to provide learners with an accessible AI-powered learning assistant that can answer questions, explain concepts simply, generate quizzes, summarize content and recommend structured learning paths. EduGenie combines a FastAPI web application with Google Gemini and a lightweight local language model.

Project Overview

Objective	The primary objective is to develop an AI-powered learning assistant that makes study support faster, simpler and more personalized through question answering, concept explanation, quiz generation, summarization and learning recommendations.
Scope	The project covers a web-based frontend, FastAPI backend endpoints, Gemini-powered generation for major learning tasks, a local LaMini-Flan-T5 model for concept explanation, and local development/testing using Uvicorn.
Target Users	Students, self-learners and learners who need quick explanations, revision support, practice questions or guidance on what to study next.

Problem Statement

Description	Learners often face difficulty understanding complex concepts, finding concise study material, revising long passages, preparing practice questions and deciding what to learn next. These tasks may require switching between multiple resources.
Impact	The lack of a focused learning workflow can increase study time and make it harder to obtain explanations and practice material at the right level. A unified assistant can provide these capabilities through one interface.

Proposed Solution

Approach	Develop a modular FastAPI application that accepts learner input from a simple HTML/CSS interface and routes it to the appropriate AI capability. Gemini handles Q&A, quizzes, summarization and learning recommendations, while LaMini-Flan-T5 supports lightweight concept explanation.
Key Features	• Question answering for learner queries. • Simplified concept explanations. • Quiz generation with multiple-choice questions. • Summarization of long educational content. • Personalized learning-path recommendations. • Unified web interface for selecting a task and viewing results.

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware		
Computing Resources	CPU/GPU resources for running the application and lightweight local model.	Standard development CPU; GPU optional for local model acceleration.
Memory	RAM required for Python environment, FastAPI and model execution.	Recommended: 8 GB RAM or higher.
Storage	Disk space for source code, Python packages and local model files.	Adequate local SSD space; exact allocation depends on model/package size.
Software		
Runtime / Frameworks	Application runtime and API framework.	Python 3.10+, FastAPI, Uvicorn.
Libraries / Models	AI integration and local language model components.	Google Gemini API integration; LaMini-Flan-T5-783M; project Python dependencies.
Development Environment	Tools for coding, testing and version control.	Git/GitHub and a Python development environment.
Data		
User Input	Questions, topics and passages submitted by learners through the interface.	Text entered by the user at runtime.
Persistent Storage	Database/storage layer for long-term learner records.	No persistent database is specified in the project description.
External Service	Cloud AI service used by the application.	Google Gemini API.

Planning Notes

The implementation is intentionally modular so that Q&A, explanation, quiz, summarization and learning-recommendation features can be developed and tested independently. The documented setup supports local execution with Uvicorn, while Gemini provides cloud-based generative AI capabilities.